

HSCTF 2020 Crypto - Extremely Complex Challenge

deuterium — 2020-06-06

Canonical HTML: [https://deuterium.github.io/WriteUps/2020/HSCTF/crypto/Extremely Complex Challenge/2020-06-06-HSCTF-2020-Extremely-Complex-Challenge.html](https://deuterium.github.io/WriteUps/2020/HSCTF/crypto/Extremely%20Complex%20Challenge/2020-06-06-HSCTF-2020-Extremely-Complex-Challenge.html)

Abstract

CTF writeup: SageMath has a modest workload in HSCTF 2020's Extremely Complex Challenge: recover an elliptic-curve parameter, then solve a decidedly small discrete log.

Extremely Complex Challenge

Challenge

75 Solves



Extremely Complex Challenge

403

ecc **crypto**

Eric has an elliptic curve defined over a Galois field with order 404993569381. A generator point (391109997465, 167359562362) is given along with a public key (209038982304, 168517698208). We also know that the curve is defined as $y^2 = x^3 + ax + b \pmod{p}$, and that b is equal to 54575449882. What is Eric's private key? Express the key as an integer in base 10. Use the flag format `flag{+private_key+}`.

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We are given two points on an Elliptic Curve, its order and parameter b . Using these quantities, we can recover parameter a .

$$\begin{aligned} y^2 &= x^3 + ax + b \pmod{p} \\ y^2 - x^3 - b &= ax \pmod{p} \end{aligned}$$

If we have two points on curve, (x_1, y_1) and (x_2, y_2) , we can get

```
(y1^2 - x1^3) - (y2^2 - x2^3) = a(x1 - x2) (mod p)
a = (x1 - x2)^-1 (y1^2 - x1^3) - (y2^2 - x2^3) (mod p)

(x1 - x2)^-1 is inverse of (x1-x2) modulo p
```

Now as we have all parameters, we can solve ECDLP easily since the sizes are too small.

```
import gmpy2

p = 404993569381
G = (391109997465, 167359562362)
P = (209038982304, 168517698208)
b = 54575449882

diff = (G[1]**2 - P[1]**2) - (G[0]**3 - P[0]**3)
diff = diff % p
a = gmpy2.invert(G[0] - P[0], p) * diff
a = a % p

E = EllipticCurve(GF(p), [a, b])

G = E(G)
P = E(P)
PK = G.discrete_log(P)
print(PK)
G*PK == P
#17683067357
```

Although I solved it using SageMath, I think it can be solved on python too using relevant libraries.

Interactive controls are omitted from this PDF. See the canonical HTML version.